A binary neural-network decoder for quantum error correction on a small FPGA

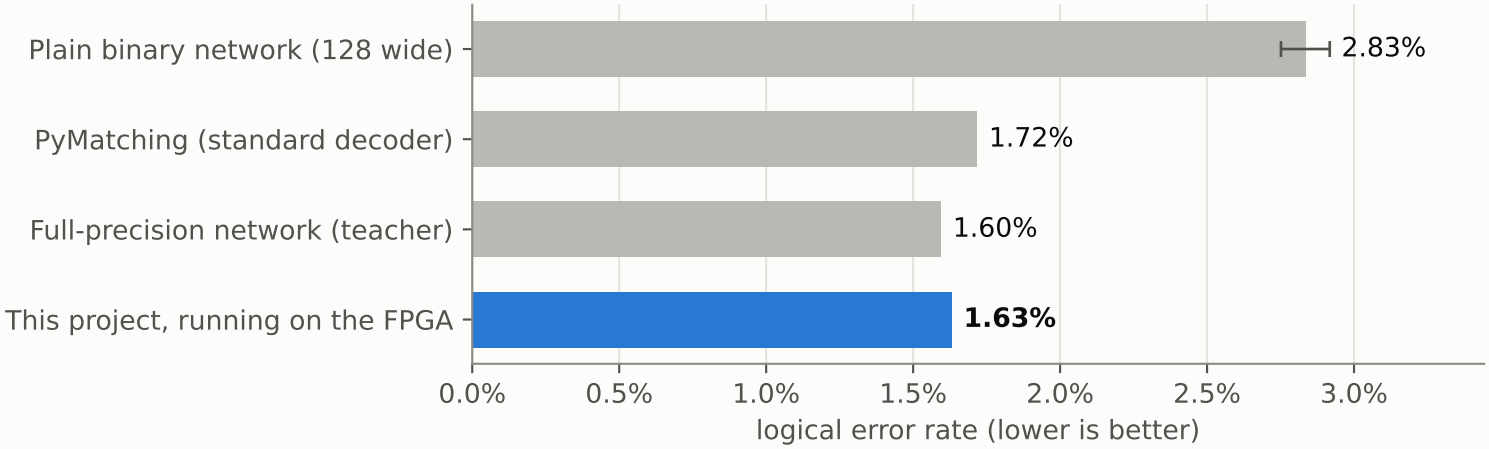
Distance-3 surface code · Stim + PyTorch + Verilog · Spartan-7 XC7S25 (PLTW S7 board)

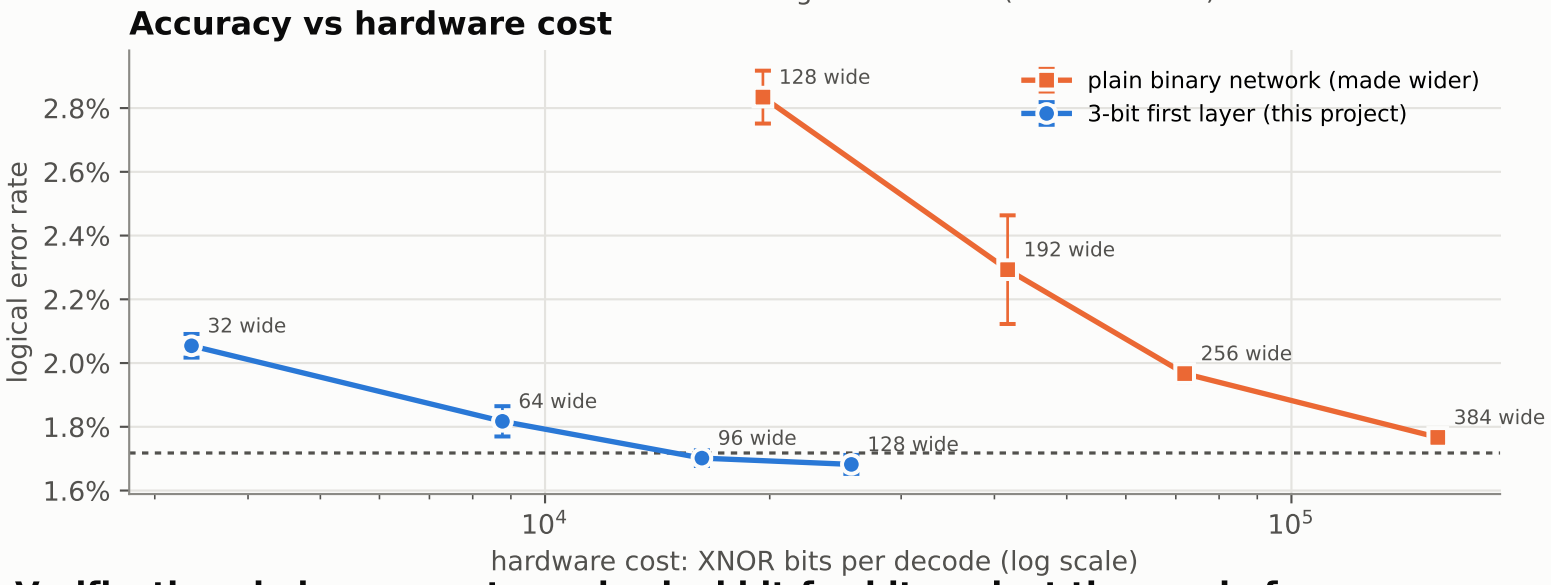
1.63%
logical error rate on the FPGA (5% below PyMatching)

0
mismatches over 1,000,000 hardware test shots

10×
fewer XNOR bits than a plain binary network of similar accuracy

11%
of the FPGA's logic (LUTs) used; timing met at 12 MHz





Verification chain: every stage checked bit-for-bit against the one before

Trained network	Integer golden model	Chip logic, simulated	Full system, simulated	Real FPGA board
PyTorch	1,000,000 shots	2,000 test vectors	774 load commands + 50 decodes over UART	1,000,000 shots
reference	**0 mismatches**	**0 mismatches**	**0 mismatches**	**0 mismatches**

What it does: a small neural network reads 24 syndrome bits from a simulated quantum memory and predicts whether a logical error occurred. Weights and activations are binary (XNOR + popcount), except a few 'salient' first-layer weights that get 3-bit precision. Choosing which detectors get the extra bits by their activity (AWQ-style) beat random choice, and expanding the first layer beat simply making the network wider by about 10× in hardware cost.